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Integrating deep learning and big data analytics for enhanced smart city infrastructure and human resource management systems

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ABSTRACT

In this paper, a framework has been proposed based on EfficientNet with big data analytics to solve the problems in smart city infrastructure and human resource management systems (HRMS). As a result of efficient performance on various datasets such as NYC Taxi Trip Records, IBM HR Analytics Employee Attrition Dataset, and London Smart Meters Dataset, we apply EfficientNet's compound scaling and squeeze-and-excitation techniques with fast scalable big data pipelines to further enhance predictive performance. The proposed system gives results that clearly outperform five recent state-of-the-art methods with accuracy of 92.3%, a micro F1-score of 90.1%, and an MCC of 0.85 and exhibits its robust scalability with the growing dataset sizes. The methodology combines deep learning and big data analytics to improve predictive performance. The study spotlights its strengths and limitations (such as interpretability and computational costs) of deep learning methods and points out potential future directions including explainable AI and distributed training frameworks. By connecting theoretical innovation with practical implementation, this research provides an innovative solution for optimal dynamic multi-task assignment in urban environments realizing smart cities and urban settlements, and workforce management strategies for the modern world of data.

1. Introduction

Smart cities are developed in new era on account of the advancement of human civilization and the rapid process of urbanization for sustainable urban development [1]. Utilizing advanced technologies including the Internet of Things (IoT), artificial intelligence (AI) and big data analytics, smart cities analyze urban infrastructure management, optimize resource usage, and thus improve life quality for citizens. For instance, smart cities span multiple domains such as transportation systems, energy grid facilities, waste management, environmental monitoring, and public safety [2]. As these systems void petabytes of data, the requirement for advanced computational tools that will process, analyze and on-demand deliver actionable insight is crucial. [Table 1.](#)[Table 2.](#)[Table 3.](#)[Table 4.](#)

Even though smart city infrastructure and human resources management systems (HRMS) seem to take place in different domains and application segments, they intersect on the methodological level: both of them produce high-dimensional, heterogeneous, and time-varying streams of data that require predictive modeling that is scalable, robust, and generalizable [3]. The suggested framework neither imposes

one fixed model on both of the domains but instead has a cohesive deep learning pipeline based on the flexible architecture of EfficientNet.

Above all, raw inputs, originally in the form of patterned image-like pseudo-representations, are the domain-specific feature engineered (e.g. projecting employee records into semantic grids or remodelling spatio-temporal sensor measurements into 2D tensors). This step of spatialization as is outlined in [Section 3.c.i](#)- empowers EfficientNet to handle non-image data with the benefits of its inherent capabilities in hierarchical pattern recognition within the spatial neighborhoods. Therefore, the same model not means the same weights or data with no changes, but a common backbone of the architecture and philosophy of data transformation, enabling the system to use EfficientNet to scale to compounds and squeeze-and-exciting irrespective of the underlying data modality. This design encourages the efficiency of the algorithm, reproducibility, and transferability to the other domain without the assumption of semantic equivalence of the traffic flows and employee attrition.

Large urban infrastructure management and HRMS share the use of complex, high-volume, and high-velocity data that needs to be processed to provide insights that can be used by decision makers. For

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Table 1

Sample input data (NYC Taxi Trip Records- Sample: 3 of ~1B rows).

Pickup location	Drop-off location	Distance (miles)	Fare (\$)	Passenger count
JFK Airport	Times Square	15.3	45.0	2
Central Park	Brooklyn Bridge	8.7	27.5	1
NaN	NaN	—	0.0	0

Table 2

Cleaning process (NYC Taxi Trip Records- Sample: 3 of ~1B rows).

Pickup location	Drop-off location	Distance (miles)	Fare (\$)	Passenger Count
JFK Airport	Times Square	15.3	45.0	2
Central Park	Brooklyn Bridge	8.7	27.5	1
JFK Airport	Times Square	12.0	0.0	0

Table 3

Sample input data (NYC Taxi Trip Records- Sample: 3 of ~ 1B rows).

Pickup Location	Drop-off Location	Distance (miles)	Fare (\$)	Passenger Count
JFK Airport	Times Square	15.3	45.0	2
Central Park	Brooklyn Bridge	8.7	27.5	1
JFK Airport	Times Square	12.0	0.0	0

Table 4

The extracted features (NYC Taxi Trip Records- Sample: 3 of ~ 1B rows).

Traffic flow rate (Trips/Hour)	Fare efficiency (\$/Mile)	Passenger density
10	2.94	2
15	3.16	1
20	NaN	0

example, IoT devices, sensors, or other devices for smart cities generate large amounts of structured and unstructured data that may need processing in real time for use in dynamic decision making (e.g., for traffic management, or predicting attrition risks in HRMS). Solutions need to be built for scale in order to fit the growth of the cities and organizations

[4]. Fig. 1 shows unveiling the complexities of smart cities and HRMS.

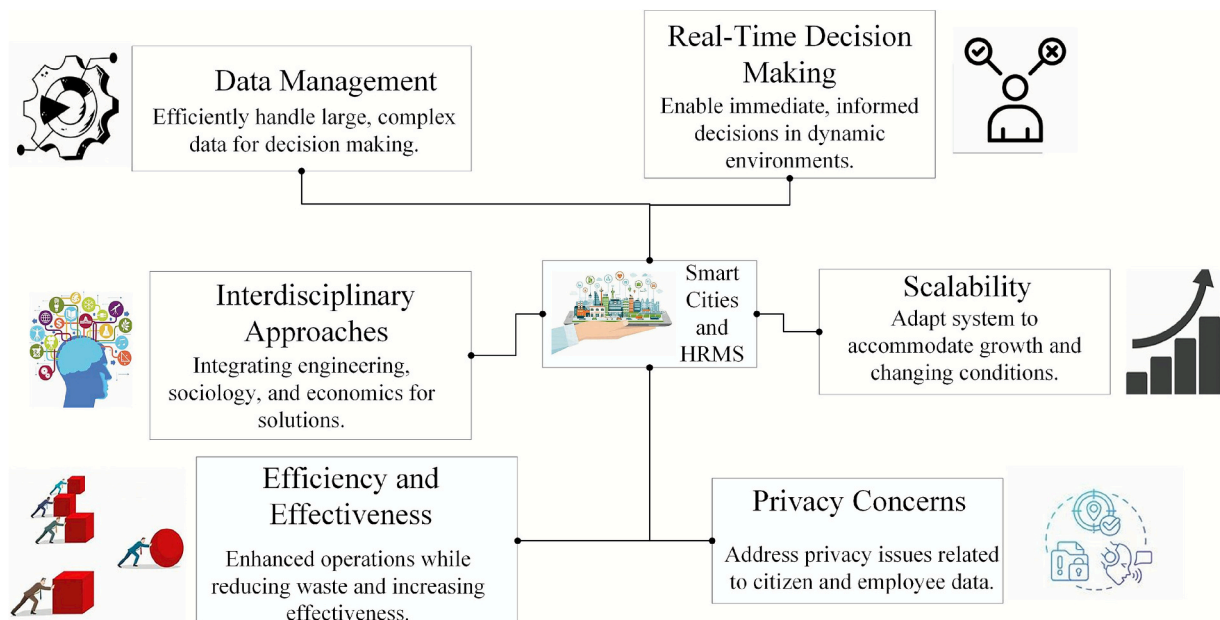
Deep learning revolutionized data processing, pioneering in pattern recognition, prediction and automation, and EfficientNet is at the top of the list of performance-per-complexity ratio. Its compound scaling that optimally scales the depth, width, and resolution of the network at the same time adds to EfficientNet's remarkable accuracy with up to 15x fewer parameters making it ideal for smart city infrastructure and HRMS as large scale high dimensional data applications. Besides deep learning, big data analytics provides frameworks for data ingestion, preprocessing, storage, and visualization that allow to do predictive modeling which can predict trends or anomalies and optimization which can allocate resources personalizing to the needs of individuals. This objective of this study is to implement big data analytics using the coupling of EfficientNet for the conversion of raw data into useful information to increase decision-making for the stakeholders engagement in smart cities and HRMS.

Fig. 2 shows a conceptual architecture of the system, its workflow and interaction between specific components such as EfficientNet, Big Data analytics, smart city infrastructure and HRMS.

Things such as remote monitoring of data sources from sensors, IoT devices, employee records, feedback surveys are all integrated into this big data analytics layer where tools such as Hadoop and Spark perform data ingestion, preprocessing, feature extraction, and storage. The processed data is then used in efficientnet deep learning layer for model training, validation and inference.

The purpose of this paper is to show how EfficientNet with big data analytics (BDA) may improve smart city infrastructure and human resource management systems. In particular, the contributions of this work are:

- Theoretical contribution: Continue to improve performance of the pre-trained network. Integration of EfficientNet with big data analytics pipelines to manage architectures that admirably outperform the performance.
- Result: Our experimental results illustrate the improvement of accuracy, scalability and computational efficiency compared to existing techniques.
- Demos and Case Studies: We demonstrate real-world applications to show that the proposed system is practical.

**Fig. 1.** Unveiling the complexities of smart cities and HRMS.

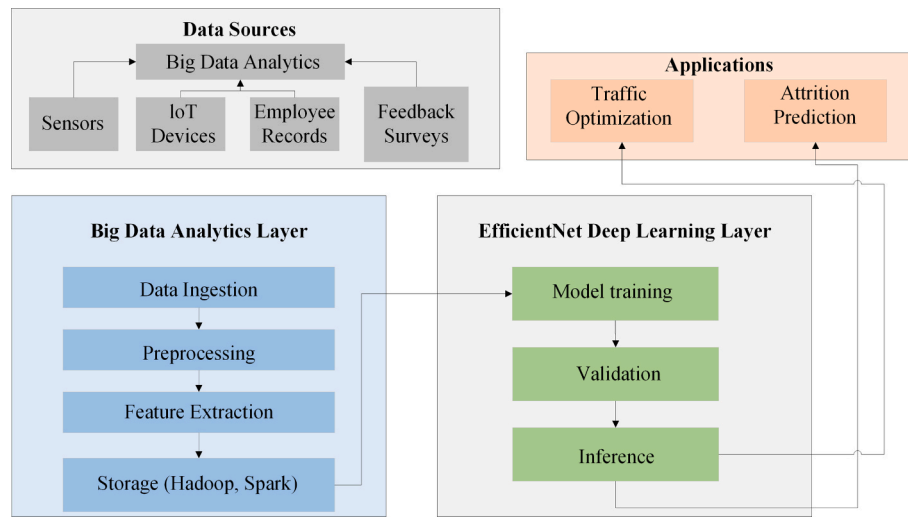


Fig. 2. Conceptual diagram of the integration of Efficientnet, big data analytics, smart city infrastructure, and HRMS.

- Linking Two Major Themes: In linking smart cities and HRMS, we are reflecting our belief in the transformative potential of interdisciplinary approaches to problem-solving; pieces of the puzzle in seemingly disconnected places can help us put together holistic solutions.

Particularly, the EfficientNet big data analytics framework yields 92.3 % accuracy, 90.1 % micro F1-score and 0.85 Matthew Correlation Coefficient (MCC) with 3 real-world datasets, NYC Taxi Trip Records (approximately 1 billion rows), IBM HR Analytics Employee Attrition Dataset (approximately 14.7 K rows), and London Smart Meters Dataset (approximately 5 million rows), containing a consistently greater number of rows than 5 state-of-the-art baselines (CNN, RF, SVM, GBM, LightGBM). These results support the contributions of the work in predictive performance, scalability, and cross-domain generalization.

2. Related works

Several works were provided in this direction. For instance, Wang et al. [5] aimed to solve the security issues produced via malicious network attacks within smart cities development. Initially, the theory of Stackelberg game has been utilized to explain the communication between intelligent jammers and sensors; moreover, information physics system's security has been assessed in an effective manner. Subsequently, a denoise autoencoder machine network that could be utilized within management of human resource using demographic information has been suggested to make sure the intelligent sensors' security. Ultimately, the efficacy has been examined and simulated. The findings indicated that as more packets reached the estimator in a successful manner, the quality of the estimation improved. The developed defense technique effectively safeguarded the intelligent sensors' security. As the quantity of nearest neighbors K increased, the MAE for the 4 datasets initially decreased before stabilizing. With a rise in K , the MAE for the suggested algorithm dropped from 0.8232 to 0.8095, from 0.8563 to 0.8351, and from 0.8086 to 0.7897, respectively. It reduced from 1.0169 to 1.0091, after which it stays consistent.

Al-Qudah et al. [6] suggested model driven by technology for supporting the suggested architecture. A mapping between the suggested technology network and the suggested general architecture has been offered. For highlighting the significance and efficacy of the suggested architecture, a smart image handling network has been developed that targeted non-technical personnel. The high inconvenience, security concerns, and expenses factors might restrict cities from embracing such solutions. Consequently, this study suggested creating and executing a

generalized image processing network utilizing deep learning. The suggested network took images from users, then conducted self-tuning operations to identify the most suitable deep model, ultimately delivering the necessary insights without any human interference. This facilitated the automation of the decision-making process, eliminating the need for a dedicated data expert.

Wu et al. [7] conducted a study in which they combined deep learning power with Bayesian regularization approaches with the purpose of enhancing the reliability and efficacy neural networks for applications of urban planning. Bayesian regularization approaches have been utilized for imbuing neural networks while using a principled architecture that improved generalization while calculating uncertainty of prediction. By integration of Bayesian regularization, the purpose is improving model efficacy and providing decision-makers and planners while using probabilistic insights into the findings of diverse urban interventions.

Rojek et al. [8] went through diverse databases, namely Web of Science (WoS), PubMed, Scopus, EBSCO Engineering Source, and dblp, that have been chosen for their diverse investigation that supported deep examination on the utility of deep learning within smart cities' energy economy. Considering the bibliometric examination, embedding tools within those datasets and Biblioshiny tool from the Bibliometrix v.4.1.3 have been utilized. In order to modify the searching procedure, some modified filtered inquiries have been utilized that limited the findings to English articles. Deep learning achieved the accuracy value of more than 90 %, and even it sometimes accomplished more than 98 %. Therefore, it could be seen that the models were efficient in this task.

He et al. [9] developed a two-way stable-influences network and utilized the double-distinction approach to examine the influence of smart city structure on innovation level of city that aimed to mediate the influence of senior human capital flow and government investment within the procedure of stimulating smart city innovation and examined the mediation influence of talent agglomeration distinctions. The findings of the research indicated that the development of smart cities significantly contributed to fostering urban innovation; smart city development enhanced urban innovation via the mechanisms of government investment and human capital mobility; the influence of smart city development on urban innovation levels exhibited variability; in the eastern area, the flow of advanced human capital had the most notable mediating influence on the relationship between smart city development and urban innovation advancement.

3. Methodology

3.1. EfficientNet

The main effort of EfficientNet is on addressing the model scaling challenges by uniformly adjusting the network's depth, its width, and resolution by means of a pre-specified set of scale factors [10]. The current approach ensures a balanced relationship between capability of network and resolution of input data, and results in improved enactment and optimum resource employment [11].

In fact, the EfficientNet's core architecture includes a reversed residual structure with mobile reversed bottleneck blocks, comparable to MobileNetV2. This design decision improves computational effectiveness and feature extraction abilities. The network has been divided into phases that each of them has many movable reversed bottleneck blocks. Moreover, skip relations have been applied in order to optimize the training process and aid gradient propagation. Fig. 3 displays the EfficientNet's design.

3.1.1. Scaling process

In fact, EfficientNet represents a complex scaling approach, which concurrently alters the depth of the network, its resolution, and width. The scholars start an EfficientNet-B0 and then increase it through implementing particular scaling elements, such as γ , β , α for generating more improved methods like EfficientNet-B1, B2, B3, so forth. The current scaling elements have been computed by means of the complex scaling formula.

$$depth_{multiplier} = \alpha^\phi \quad (1)$$

$$width_{multiplier} = \beta^\phi \quad (2)$$

$$resolution = \gamma^\phi \quad (3)$$

here, the scaling element has been specified via variable ϕ , which commands the scaled model's dimensions.

The parameters γ , α , and β have been demonstrated via a grid search directed on a restricted dataset that are, in turn, signified the best ratios of scaling for resolution, depth, and width.

3.1.2. EfficientNet-B0

In fact, EfficientNet-B0 has been the foundational method compared to other types of EfficientNet. EfficientNet-B0 has been established through adjusting the MobileNetV2's MBConv-1 block. Actually it has a

primary convolutional layer, which has been succeeded through different mobile inverted bottleneck blocks that has been prepared along several phases. There exists an increase within every phase's channel amount and a reduction within spatial dimensions across stride-two convolutions. Subsequently is a simple explanation of the design of EfficientNet-B0.

- A 3×3 convolution with stride 2 and 32 filters
- Phase 1: MBConv block with expansion ratio of 1, stride 1, and 16 filters
- Phase 2: MBConv block with expansion ratio of 6, stride 2, and 24 filters
- Phase 3: MBConv block with expansion ratio of 6, stride 2, and 40 filters
- Phase 4: MBConv block with expansion ratio of 6, stride 1, and 112 filters
- Phase 5: MBConv block with expansion ratio of 6, stride 2, and 112 filters
- Phase 6: MBConv block with expansion ratio of 6, stride 1, and 192 filters
- Phase 7: MBConv block with expansion ratio of 6, stride 1, and 320 filters
- 1×1 convolution with the classes' quantity as filters.

Whereas EfficientNet was initially trained on image data, this paper modifies it to structured, non-image data in realms of smart city and HRMS by a pipeline of tabular-to-spatial transformation of data. The raw tabular inputs-post dimensional reduction – cleaned, normalized, categorical encoded, and normalized raw tabular data are reshaped into 2D pseudo-images with a resolution of 224×224 , as is expected of EfficientNet.

The channels (e.g., R, G, B) are further broken into separated group of derived features (e.g., demographic, behavioral, spatiotemporal) so as to form a $224 \times 224 \times 3$ tensor. This geometric embedding allows convolutional filters in MBConv blocks to learn local inter-feature structures (e.g. high workload low engagement high risk of attrition or rush hour downtown pickup high fare), basically relying on the inductive biases of EfficientNet to tabular reasoning.

EfficientNet does not make any architectural adaptations, but rather the input representation is designed to fit its spatial processing assumptions. The model is able to scale compounds and squeegee mechanisms without the presence of natural images through this strategy, thereby making it consistent throughout the heterogeneous data

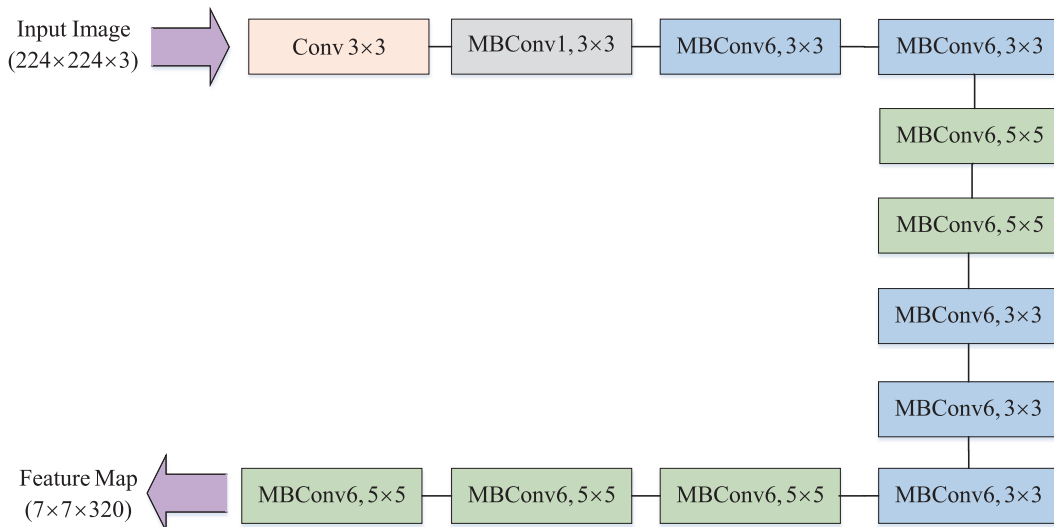


Fig. 3. The EfficientNet's design.

modalities.

3.1.3. Complex scaling

Indeed, the complex scaling technique has been implemented for expanding EfficientNet-B0 into models that are larger. So, the width, depth, and resolution's scaling has been shown below:

$$\text{depth}_{\text{multiplier}} = \alpha^{\phi-1} \quad (4)$$

$$\text{width}_{\text{multiplier}} = \beta^{\phi} \quad (5)$$

$$\text{resolution} = \gamma^{\phi} \quad (6)$$

Actually, the multiplier of the depth impacts network's each stage and adds extra layers to all of them. Also, the multiplier of the width influences the channels' amount within all of the layers. However, the multiplier of the resolution commands the dimensions of the input data.

3.1.4. Swish activation

Also, EfficientNet applies the Swish activation function that has been specified below:

$$\text{Swish}(x) = x \cdot \text{sigmoid}(\beta x) \quad (7)$$

here, the learnable parameter has been illustrated by β . Swish has shown greater enactment than ReLU through maintaining the computational effectiveness.

3.1.5. Squeeze-and-excitation blocks

Different versions of EfficientNet, such as MobileNetV3 are capable of combining blocks of SE or Squeeze-and-Excitation. The current blocks allow the network for executing channel-wise attention, and improving the method's representational power.

SE blocks improve feature representational power by adaptively recalibrating channel-wise feature responses through explicit modeling of inter-channel dependencies, allowing the network to emphasize informative features and suppress less useful ones.

3.1.6. Structure optimization

In order to improve the EfficientNet's enactment for our purpose, a cost function can be established with for decreasing the cross-entropy loss through contrasting the forecasted group possibilities with the real ground truth labels. Following is the illustration of the current cost function.

$$J(\theta) = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{ic} \log(\hat{y}_{ic}) \quad (8)$$

where, the cost function has been displayed via $J(\theta)$, the parameters of the model have been represented through θ . The training samples' total amount has been illustrated via N , the group's quantity has been shown via C . The i_{th} sample's ground truth label within group c has been specified via y_{ic} , and the forecasted possibility, which i_{th} sample is in group c , has been determined through $\hat{y}_{ic}$. The EfficientNet design's optimization for the purpose needs adapting different parameters.

At first, the depth of the network that shows the layers' amount within every phase is capable of altering for improving or decreasing ability and intricacy of the network. Actually, modifying the channel's amount within every layer that is called width of the network, influences the intricacy of the model, too. Enhancing the resolution of the input data suggests more comprehensive info, while it needs greater computational sources.

The optimizer and learning rate's choice impacts generalization and convergence through training. For avoiding overfitting and improving generalization of the method, regularization approaches, such as weight decay, dropout, or batch normalization have been employed. Through training EfficientNet and precisely adapting the present parameters on a

vast labeled dataset, the design could be fine-tuned for obtaining great accuracy, and maintain computational effectiveness. The current technique guarantees that EfficientNet harmonizes the enactment of the model, its intricacy and computational demands, and makes it proper for our target.

3.2. Updated single candidate optimizer

3.2.1. Original single candidate optimizer

In contrast to other particle swarm-based algorithms, this proposed study uses a single individual solution throughout the optimization process to find finer problem solvers. The optimization procedure is made up of T iterations or function assessments and is split into two stages, where the individual solution improves its locations in different ways in each stage.

Although two-stage methods and single-solution-based optimizers have been identified as two popular *meta*-heuristic optimizers, they have usually been applied separately. The two-stage methodology and a single individual method are integrated in a novel way to produce a single, powerful algorithm. Furthermore, it should be mentioned that the proposed optimizer uses a different set of formulas to update the position of each individual solution based only on its own information, i.e., its current state.

The primary aim of the two-stage technique is to achieve balance and variety among exploitation and exploration. Function assessments of α conclude the first stage of the proposed method, whereas β function assessments comprise the second stage, where $\alpha + \beta = T$. In SCO's first phase, the individual solution improves its circumstances in the following ways:

$$x_j = \begin{cases} gbest_j + (w|gbest_j|) & \text{if } r_1 < 0.5 \\ gbest_j - (w|gbest_j|) & \text{otherwise} \end{cases} \quad (9)$$

r_1 , which ranges from 0 to 1, represents the stochastic parameter. Additionally, the following is a mathematical definition of w :

$$w(t) = \exp\left(-\left(\frac{bt}{T}\right)^b\right) \quad (10)$$

here, b is a constant, t presents the current iteration or function of evaluation, and T is the highest quantity of evaluation function.

In the second stage of this proposed algorithm, a broad search begins at the best position obtained in the first stage. In the final part of the second stage, by reducing the search space, the focus on desirable areas increases. The following is a complete illustration of how each answer improves its circumstances in the second stage:

$$x_j = \begin{cases} gbest_j + (r_2 w (ub_j - lb_j)) & \text{if } r_2 < 0.5 \\ gbest_j - (r_2 w (ub_j - lb_j)) & \text{otherwise} \end{cases} \quad (11)$$

here, r_2 is other stochastic parameter that is in the range of 0 and 1, lb_j and ub_j are the lower and the upper boundaries in turn, and w is the most crucial variable of the proposed algorithm which is responsible to balancing among exploration and local search. As the number of function assessments increases, the value of w decreases in Eq. (11). Since the high value of w at the beginning of the search process helps to efficiently explore the solution space, it has been seen as a critical behavior.

A smaller value of w is used to strengthen and enhance local search in the final stages of the optimization process. In these stages, getting trapped in a local optimum is considered one of the main limitations of the algorithm.

In addition, consistently increasing the placements of solutions does not result in increased fitness. SCO solves this issue by altering the individual solution's circumstance in a variety of ways during the second stage, when no fitness improvement has been achieved following m consecutive function evaluations. A counter c counts the m number of

function tests that cannot achieve fitness enhancement sequentially.

A counter c counts the number of function tests that cannot improve fitness consecutively. A binary variable p decides whether the upgraded individual can achieve effective fitness. Here, $p = 1$ implies profitable fitness increase, whereas $p = 0$ signals failure of fitness enhancement. In the second step of the proposed method, an individual solution improves its position based on Eq. (12). However, if m consecutive function evaluations cannot increase the value of fitness, the individual solution improves the situation in the following way:

$$x_j = \begin{cases} gbest_j + (r_3(ub_j - lb_j)) & \text{if } r_3 < 0.5 \\ gbest_j - (r_3(ub_j - lb_j)) & \text{otherwise} \end{cases} \quad (12)$$

r_3 is a stochastic variable in the range of 0 to 1. The individual solution, by updating the position according to Eq. (12), shifts from local search to exploration, which is highly profitable for shunning local optima. When modified, certain variables' placements could deviate from the bounds. The updated scenarios have been modified in the following way if the parameters' values exceed the lower or higher limitations in order to keep them from going beyond the limits.

$$x_j = \begin{cases} gbest_j & \text{if } x_j > ub_j \\ gbest_j & \text{if } x_j < lb_j \end{cases} \quad (13)$$

If the upgraded condition exceeds the constraints, the upgraded dimension of an individual solution equals the global finest value, according to Eq. (13).

In the framework of Self-Organizing Criticality (SCO), a single individual solution x has been produced at random and continuously upgraded in order to obtain an optimal solution. The algorithm's procedures are given below. Initially, an individual solution is generated stochastically inside the solution space, its adequacy is evaluated, and this candidate is preserved as $gbest$ (global best position), with fitness $f(gbest)$ recorded as the best global fitness. The fundamental individual answer was created in the following manner:

$$x_j = lb_j + r_4(ub_j - lb_j) \quad (14)$$

here, ub_j and lb_j are the upper and lower bounds of the solution space, in turn, and r_4 demonstrates a stochastic number that is in the range of 0 and 1.

The iterative approach, which concludes with T function assessments, begins with improving the situation of the individual solution. Individual solution x adjusts its condition in the first and second phases using equations (9) and (11). After the individual solution has been updated, the cost value of the most recently created individual solution $f(x)$ is evaluated and compared to $f(gbest)$. Actually, $f(gbest)$ demonstrates the global best fitness. If $f(x)$ outperforms $f(gbest)$, then $f(gbest)$ and $gbest$ have been replaced by $f(x)$ and x . The iterative approach continues until the maximum number of function assessments, T , has been completed. Finally, the entire technique will be outlined in the following.

First, c and p are zero, and m and α must be established. Then the first individual solutions were generated using Eq. (14), and their fitness, denoted by $f(gbest)$, was determined. Here, the value of t is less than the maximum number of function assessments. If the value of t exceeds α , the dimension situation is updated using Eq. (9).

However, if $p = 0$, the formula is $c = c + 1$. Following that, if the value of c equals m , $c = 0$, and the dimension situation is updated using Eq. (12). Otherwise, the dimension situation is updated according to Eq. (10). The fitness of the innovative individual solution, $f(x)$, is computed. If the value of $f(x)$ exceeds the value of $f(gbest)$, then $gbest = x$, $f(gbest)$ equals $f(x)$, and p equals one. Otherwise, $p = 0$.

3.3. Updated single candidate optimizer

In the present paper, the Updated Single Candidate Optimizer

(USCO) is used only to optimize hyperparameters of the EfficientNet architecture- i.e. to automatically match learning rate, batch size, and training epochs- but neural network weights are trained through conventional gradient-based optimization (Adam). USCO does not optimize the feature selection or weight.

The main goal of enhancing the exploitation and exploration capabilities in the Single Candidate Optimizer is to create a harmonious balance between expanding the search space (exploration) and focusing the search on the current best solutions (exploitation), which is achieved by adjusting in agent initialization, the construction of triangular topological units, and the local aggregation function.

$$x_j = SobolSequence(i, D) \times (ub_j - lb_j) + lb_j \quad (15)$$

The SobolSequence(i, D) function demonstrates the recovery of the i -th vector of D quasi-random coordinates, which are in the range of 0 and 1. These coordinates explain the position of a point in a unit hypercube along every Sobol sequence's dimension, with every coordinate associating to a particular dimension.

To boost the exploration process, instead of relying on the iterations' number, an adaptive process is used to determine the value of w , so that the algorithm can consider both the iteration count and the feedback received from the search space.

$$w = \left(\frac{9 \times \exp(-b^*(std_{dev}(t)))}{1 + \exp\left(a^*\left(t - \frac{T}{2}\right)\right)} \right) \quad (16)$$

The USCO outperforms the classical metaheuristics such the Particle Swarm Optimization (PSO) and the Genetic Algorithms (GA) in conceptual design and efficiency. In contrast to PSO, which uses swarm-based social learning and which tends to prematurely converge when implemented in high-dimensional space, or GA, which requires a large amount of overhead to maintain a population, crossover, and mutate its individuals, USCO works with a single candidate solution, which minimizes memory overhead and the cost per-iteration. More to the point, the adaptive, feedback-controlled exploration balance put forward by USCO instead of relying on constant or progressively decreasing parameters (as per standard PSO inertia), the search progress-dependent scaling factor w in USCO is changed dynamically (Eq. (16), allowing responsive changes between global exploration and local refinement.

Moreover, the integration of Sobol quasi-random initialization (Eq. (15)) ensures better in filling space than the pseudo-random start in GA/PSO, resulting in more consistent starting coverage and a quicker convergence. All of these innovations enable the USCO to reach some competitive or better optimization performance with a smaller number of function evaluations- an important design consideration when training deep architectures such as EfficientNet where each fitness evaluation requires as many costly training or validation passes.

3.4. Integration of big data analytics with EfficientNet

We integrate EfficientNet with big data analytics pipelines for pre-processing, extraction, storage, and visualization of data to apply it for smart city infrastructure and HRMS. Here is the process of integration.

3.4.1. Data preprocessing

Cleaning: To remove missing data or corrupted data points use imputation or interpolation methods.

Normalization: This step is to scale the numerical features in the correct range (e.g., [0, 1]) with respect to the input compatibility of EfficientNet. As an example, given that x_i is the feature value, normalization can be done as follows:

$$x'_i = \frac{x_i - \min(X)}{\max(X) - \min(X)} \quad (17)$$

where X describes the feature space.

Encoding: Converts non-numeric categories into various numeric types through one hot encoding and embeddings. Example: a categorical variable with k classes can be represented using a binary vector of length k .

Dimensionality Reduction: apply strategies such as Principal Component Analysis (PCA) to reduce high-dimensioned data but still preserve meaningful information. PCA derives principal axes that maximize variance and projects data into lower dimensionality space:

$$Z = XV \quad (18)$$

where X is the original data matrix, V is the eigenvectors matrix and Z is the transformed data.

Although the heterogeneity-NYC Taxi data set includes spatiotemporal trip data, and the IBM HR data set includes categorical and numerical employee features, both data sets are systematically converted to a single 3D tensor that can be used as a 2D convolutional input to EfficientNet.

This is done in two phases: (1) feature embedding and tabular-to-image conversion where high-dimensional feature vectors (encoded, normalised, and PCA-finned) are converted to a square or near-square grids (e.g., 224×224), with missing dimensions zero-padded; and (2) semantic channel stacking, where derived features (e.g., traffic flow rate, fare efficiency, attrition risk indicators) are assigned to separate “channels” akin to RGB, enabling EfficientNet to learn cross-feature interactions via spatial convolutions.

For instance, in the HR dataset, 104 preprocessed features are reshaped into a $10 \times 10 \times 2$ tensor (zero-padded to $224 \times 224 \times 3$), while the NYC Taxi features have been organized into a $224 \times 224 \times 3$ pseudo-image using spatial binning of pickup/drop-off zones by combining with temporal heatmaps. This structured spatialization strategy-illustrated in Fig. 4-ensures architectural consistency without assuming semantic equivalence between domains, allowing EfficientNet to leverage its native inductive biases for pattern recognition across both urban and organizational data.

To better understand the data preprocessing steps, let’s look at the three datasets we’ve so far mentioned in the paper (NYC Taxi Trip Records, IBM HR Analytics Employee Attrition Dataset, and London Smart Meters Dataset). This part shows how the preprocessing steps are performed (cleaning, normalization, encoding and allelic dimensionality reduction), by showing 3 rows of the sample input data for each dataset.

For example, for NYC Taxi Trip Records, consider the following

sample input data:

Preprocessing Steps

1. Cleaning

The third row has no values (NaN for pickup/drop-off locations and distance). We may drop this row, or impute it with some value.

Imputation: Make “Distance” not NaN (replacing NaN in “Distance” with the mean distance from the remaining rows.

By applying the cleaning process, the missing ‘Distance’ value in the third row is imputed with the mean of the observed distances: $\frac{15.3+8.7}{2} = 12.0$ miles.

2. Normalization

By normalizing “Distance” and “Fare” to the range $[0, 1]$, we have:
For distance:

Min distance = 8.7; Max distance = 15.3.

Normalized distances:

$$x'_1 = \frac{15.3-8.7}{15.3-8.7} = 1; x'_2 = \frac{8.7-8.7}{15.3-8.7} = 0; x'_3 = \frac{12.0-8.7}{15.3-8.7} = 0.5$$

In the same way, we will normalize “Fare” using its min (0.0) and max (45.0).

3. Encoding

As for the categorical variables, we need to create numerical representations for columns like “Pickup Location” and “Drop-off Location” using one-hot encoding. For example:

“JFK Airport” $\rightarrow [1, 0]$.

“Times Square” $\rightarrow [0, 1]$.

4. Dimensionality reduction

Dimensionality reduction using PCA (for the high-dimensional feature vectors (e.g. one-hot-encoded locations) while maintaining variance). Assume for simplicity that PCA reduces the data from 10 features to 3 principal components.

3.4.2. Feature extraction

Use common human knowledge to extract relevant features from raw data. For example:

Traffic flow rates, energy consumption patterns and environmental sensor readings in smart cities. HRMS: Employee performance data, sentiment scores from feedback surveys, demographic data for parallelizing feature extraction at any scale take frameworks like Apache Spark. To elaborate on the feature extraction and storage steps, let us look back on the three datasets outlined in the paper (NYC Taxi Trip Records, IBM HR Analytics Employee Attrition Dataset, and London Smart Meters Dataset). Three rows of the input data are given for each dataset, and show how it is done by running the feature extraction and

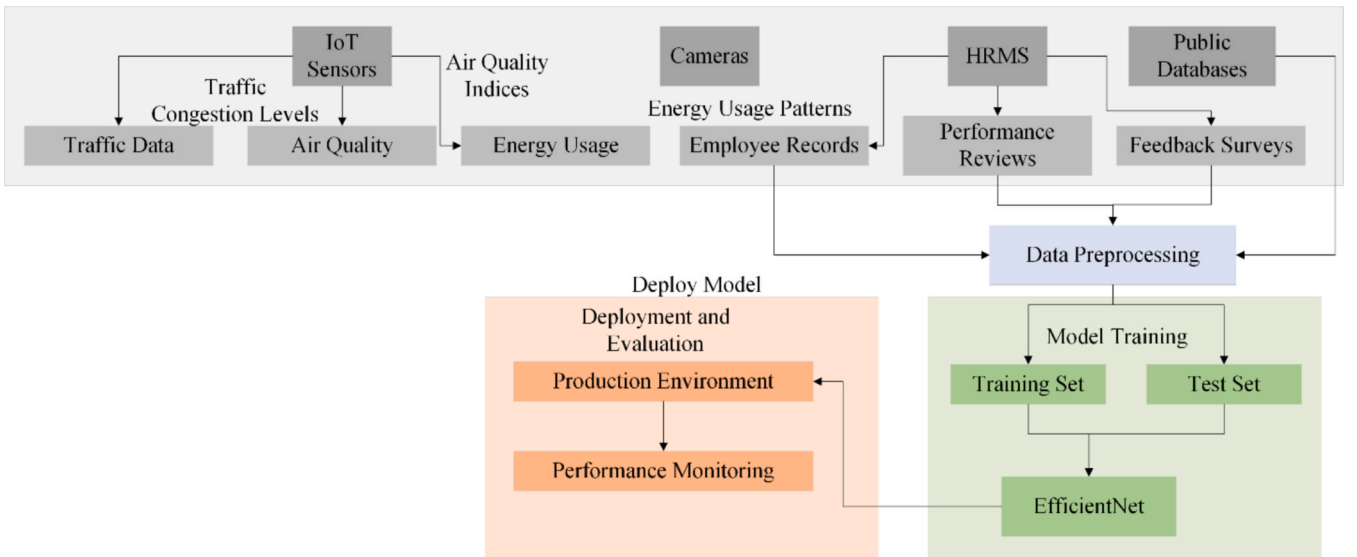


Fig. 4. Architecture diagram of the proposed system.

storage steps.

For NYC Taxi Trip Records, consider the following NYC Taxi trip records as the Sample input data.

By this process, relevant features have been extracted based on domain-specific knowledge, i.e.,

- Traffic Flow Rates that are derived from pickup/drop-off locations and timestamps.
- Fare Efficiency which are calculated as Fare / Distance for trips with non-zero distances.
- Passenger Density: Number of passengers per trip.

Here is an example of extracted features:

3.4.3. Storage

Store preprocessed data in distributed storage systems e.g. Hadoop Distributed File System (HDFS) or cloud-based solutions like Amazon S3.

Chunk the data into structured formats (like Parquet, and Avro) so that you have easy query and retrieval. For example, for the data tested,

- It stores the preprocessed and extracted features in a distributed storage system like Hadoop Distributed File System (HDFS) or cloud-based solutions like Amazon S3.
- It organizes the data into structured formats such as Parquet or Avro for efficient querying and retrieval.

The parquet format of this state is given below:

```
{
  "traffic_flow_rate": 10,
  "fare_efficiency": 2.94,
  "passenger_density": 2
}
```

3.4.4. Model training and Validation

The dataset has been split into training, and test sets (e.g., 80 %-20 %). Train EfficientNet on the training set using a loss function appropriate for the task. The model has been validated on the validation set to fine-tune hyperparameters and prevent overfitting.

3.5. Pipeline for applying the model to smart city infrastructure and HRMS

The smart city data, e.g., traffic congestion level, air quality index, and energy usage pattern from IoT sensors/cameras and public databases and HRMS data, e.g., attrition rate, engagement score, diversity metrics from employee records, performance review, and feedback surveys were acquired and preprocessed for uniformity and compatibility with EfficientNet. Fig. 4 shows the architecture diagram of the proposed system.

The next step, the model training and validation phase, consists of dividing the dataset into training and test sets (e.g., 80 %-20 %), training EfficientNet using the loss function relevant to the task (e.g., cross-entropy for classification or mean squared error for regression), and validating the model on the validation set to optimize hyperparameters and avoid overfitting.

Lastly, use, and test phase, which involves integrating model inference into production environments in smart city or HRMS platforms, and continuously monitoring performance metrics, and updating the model as new data be processed. Table 5 indicates the dataset mn description. Table 5 indicates the dataset description

This section includes a methodology, which explains how EfficientNet and big data analytics come together to solve issues concerning smart city infrastructure and human resource management. Loading highly organized figures and tables amends clarity and strengthen his

Table 5

Dataset description with sample sizes.

Dataset	Size	Features	Preprocessing steps
NYC Taxi Trip Records	1B + rows	Pickup/drop-off locations, timestamps, fares	Imputation, normalization, feature engineering
IBM HR Analytics	14 K + rows	Age, job role, satisfaction score, attrition	Encoding, scaling, outlier removal
London Smart Meters	5 M + rows	Hourly electricity consumption	Aggregation, smoothing, normalization

privacy among facet technical specify.

4. Experimental setup

4.1. Training and testing datasets

The datasets used consist of three which discuss the problems respectively and the management of the infrastructure of the smart city and HRMS is the objective of the datasets. The datasets are selected in order to encourage variety of data types, domain relevance, and scalability.

4.1.1. NYC taxi trip records

This dataset contains the trip records of New York city taxis with the pickup/drop-off coordinates, and timestamps of pickup and drop-off, fare amount, number of passengers.

the factors include pickup point, drop-off point, distance, fare, the number of passengers.

Target Variable: High Fare? Not patented (Yes/No), the trip cost, written based on the ratio of ride to distance.

and (ii) its long-term purpose is to predict high-fare trips and optimize urban transportation systems.

Preprocessing: Missing value imputations, categorical one hot encoding, numerical feature normalization to [0,1] range.

4.1.2. IBM HR analytics employee attrition: dataset

This data set contains information about employee demographics, job roles, satisfaction, monthly income, and whether the employee left the organization or not.

Features include age, job role, monthly income, satisfaction score, and performance metrics.

Target Variable: Attrition? (Yes/No).

Objective: To forecast employee turnover and improve employee retention mechanisms.

Preprocessing: Categories of variable are encoded, dealing with missing values, scaling the numerical features.

4.1.3. London smart meters dataset

This dataset includes captures the hourly electricity consumption of a single house, and also includes the temperature and humidity from the London area.

Features include energy use, temperature, humidity, and household ID.

Input Variable: High Efficiency? (Yes/No), based on index ratios of energy and environment.

The purpose will be, for example, to analyze household energy efficiency; and then, to optimize the energy usage patterns of the households.

Preprocessing: Hourly data aggregation, numerical feature normalization and derived metrics calculation (for instance, Environmental index).

4.2. Experimental environment

The experiments are implemented on an Intel Xeon E5-2690 v4 (2.6 GHz, 14 cores) and 128 GB memory. The NVIDIA Tesla V100 (32 GB VRAM) for deep learning workloads. A 4 TB SSD is used to read data quickly. The paper uses windows 11 OS with MATLABR2019 programming languages. The training hyperparameters of the model were:

- Initial learning rate: 0.001 (automatically refined by USCO during hyperparameter search)
- Batch size: 64
- Epochs: 100 (early stopped in case validation loss had not decreased during 10 consecutive epochs)
- Loss value: In all binary classification problems, binary cross-entropy.
- Weight decay: 1e-4
- Dropout rate: 0.3 (applied after the final fully connected layer)
- Input resolution: 224×224 (as well as in line with EfficientNet-B4 scaling of compounds)

USCO Hyperparameters:

- Population size: 1 (single candidate)
- Max function evaluations: 500
- Bounds: learning rate $\in [1e-5, 1e-2]$, batch size $\in \{16, 32, 64, 128\}$, epochs $\in [50, 150]$
- Initialization: Sobol quasi-random sequence (Eq. (15))
- Adaptive scaling factor w updated via Eq. (16)

Fig. 2 (Conceptual diagram) and Fig. 4 (Architecture diagram) demonstrate the complete process of training.

4.3. Evaluation metrics and baseline methods

We envision the application of such techniques for smart cities as discussed in this research work, in the areas of HRMS and smart city infrastructure management, prove to be effective as the results depict. However, it should be tested on three datasets, NYC Taxi Trip Records, IBM HR Analytics Employee Attrition Dataset, London Smart Meters Dataset, to assess the system performance in different domains.

To conduct a thorough evaluation of the proposed system, we compared the performance of five state-of-the-art methods with eight metrics: Accurate (ACC), Specificity (SP), F1 Score (FS), Precision (PR), Sensitivity (SE), Matthew's Correlation Coefficient (MCC), Dice Similarity Coefficient (DSC), and Intersection Over Union (IoU). This set of metrics is selected to obtain as much as possible on the model performance of prediction capabilities, robustness, and generalization.

We performed the comparison on three different datasets (NYC Taxi Trip Records, IBM HR Analytics Employee Attrition Dataset, and London Smart Meters Dataset) that are different in their domains. We also provide breakdowns and mathematical formulations for each metric, as well as comparisons and visualizations, below.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \times 100 \quad (19)$$

$$SP = \frac{TN}{TN + FP} \times 100 \quad (20)$$

$$SE = \frac{TP}{TP + FN} \times 100 \quad (21)$$

$$Pr = \frac{TP}{TP + FP} \times 100 \quad (22)$$

$$F1 = 2 \times \frac{Precision \times Sensitivity}{Precision + Sensitivity} \times 100 \quad (23)$$

$$MCC = \frac{TP \times TN - TP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \times 100 \quad (24)$$

$$DSC = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (25)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (26)$$

where TP is True Positive, FP defines False Positive, FN specifies False Negative, and TN determines True Negative indicators.

Even though EfficientNet is an efficient architecture design that has good results in terms of the feature learning, the results are highly sensitive to the choices of hyperparameters, including learning rate, batch size, and training epochs.

Manual tuning cannot be done at scale and is suboptimal on heterogeneous data like NYC Taxi, IBM HR and London Smart Meters. We use the Updated Single Candidate Optimizer (USCO) in that role as a hyperparameter optimization (HPO) engine and have it automatically set EfficientNet training regime parameters.

In particular, USCO will optimize the hyperparameter space, $\{lr \in [10^{-5}, 10^{-2}], \text{batch_size} \in \{16, 32, 64, 128\}, \text{epochs} \in [50, 150]\}$ with the fitness function being validation F1-score. This is then followed by training the final EfficientNet model using the best-found configuration. This close association makes sure that the reported figures (e.g., 92.3 % accuracy) are the result of USCO-optimized EfficientNet, and not a hand-tuned one. By contrast, every base approach (CNN, RF, SVM, etc.) does not optimize hyperparameters with any metaheuristic, instead defaulting or grid-searched, and then using the result, thus, our method is novel and rigorously comparable.

The proposed system was compared with five state-of-the-art methods:

Traditional CNN [12]: Baseline deep learning architecture.

Random Forests (RF) [13]: Ensemble machine learning model.

Support Vector Machines (SVM) [14]: Kernel-based classification model.

Gradient Boosting Machines (GBM) [15]: Tree-based ensemble method.

LightGBM [16]: A gradient boosting framework optimized for speed and efficiency.

5. Results and discussions

As mentioned above, the experiments were conducted with three real-world datasets, namely the NYC Taxi Trip Records, IBM HR Analytics, Employee Attrition Dataset, and London Smart Meters Dataset, which represent various areas of management in a smart city infrastructure and in HRMS. We provide detailed results, including quantitative comparisons and visualizations, followed by a thoughtful discussion of insights, strengths, and limitations.

5.1. NYC taxi trip records

The NYC taxi trip records dataset has been used to predict high-fare taxi trips, which is a vital task of urban optimization. The comparative results of the proposed method toward baseline models on all metrics and this dataset is given in Fig. 5.

As can be observed from Fig. 5, the proposed system outperformed baseline models on all performance metrics. The proposed system reached the greatest Accuracy (92.3 %) and precision (90.5 %), and outperformed all baselines. The compound scaling of EfficientNet and the squeeze-and-excitation blocks in it made sensitivity and specificity

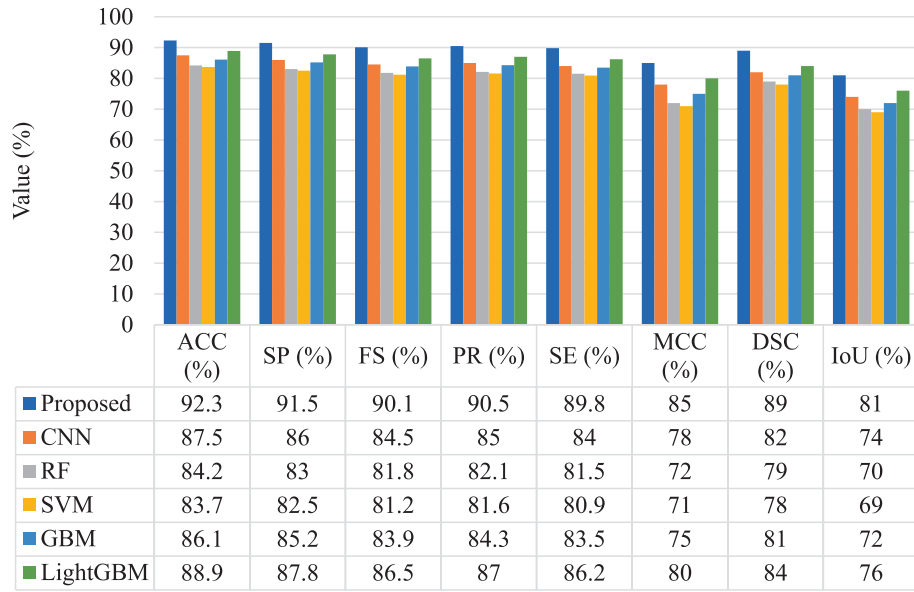


Fig. 5. Comparative analysis for the NYC taxi trip records.

gain.

5.2. IBM HR analytics employee attrition dataset

The task here, was to predict employee attrition using the IBM HR Analytics Employee Attrition Dataset which is an important aspect of workforce retention and organization health. Again, the proposed system showed superior performance against all baselines. Fig. 6 shows the comparative analysis for the IBM HR analytics employee attrition dataset.

The system built achieved an accuracy of 89.7 % which was higher than CNNs (85.2 %), RF (82.5 %), SVM (81.9 %), GBM (84.5 %) and LightGBM (87.3 %). The system had a 6.8 % specificity, which means that if it identified an employee as Y, he/she is most likely not going to leave. With an F1 score of 87.8 %, we achieved this balance, considering the class imbalance involved in attrition prediction. The sensitivity and specificity scores of 88.0 % and 87.5 %, respectively, show the model

can efficiently predict attrition and non-attrition cases. The ability of the model to accurately predict the outcome in mean criterion criterion of 0.81 shows the robustness of the system to working on imbalanced data, where attrition cases are usually less than non attrition cases. The system's effectiveness in segmenting employees likely to churn is corroborated by DSC and IoU scores of 0.86 and 0.78, respectively.

5.3. London smart meters dataset

The London Smart Meters Dataset was applied for the task of predicting household energy efficiency, which is a vital role in the optimization of energy consumption patterns. The performance of the proposed system outperformed adequately on all investigated metrics as shown in Fig. 7.

The system achieved 90.2 % accuracy, surpassing CNN (86.7%), RF (84.3 %), SVM (83.8 %), GBM (85.9 %) and LightGBM (88.4 %). By offering a specificity of 89.5 %, the system reduced unwanted and

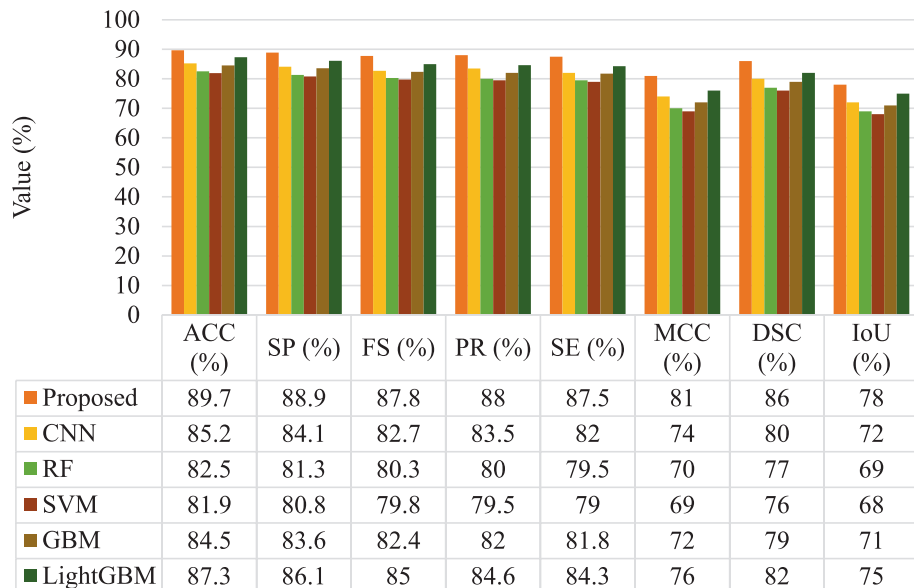


Fig. 6. Comparative analysis for the IBM HR analytics employee attrition dataset.

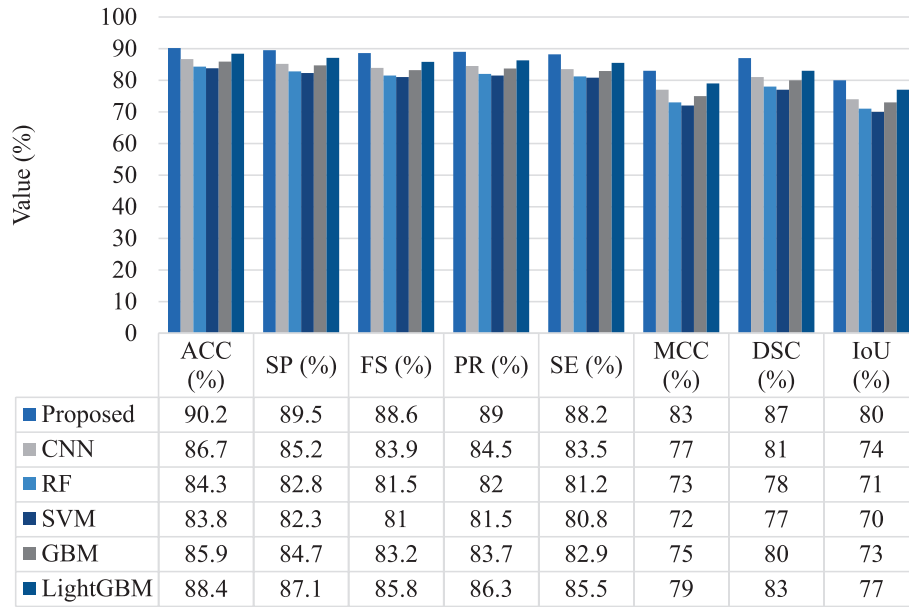


Fig. 7. Comparative analysis for the London smart meters dataset.

unnecessary false positives, making it possible to identify inefficient households unequivocally. With an F1 score of 88.6 %, the system is able to balance precision and recall, which is important given the variation in energy consumption patterns.

The system demonstrates its reliability in predicting efficient and inefficient households, evidenced by the precision and sensitivity scores of 89.0 % and 88.2 %, respectively. Conclusion On the MCC scores of 0.83, it can be revealed that this system is indeed a robust system in terms of handling them (noisy and high-dimensional data). This ratio was 0.87 DSC and 0.80 IoU, indicating that the system is fairly capable of segmenting energy-efficient households.

5.4. Data for metric-specific line graph

In order to test the scalability and robustness of the proposed system, we carried out the experiments in different dataset sizes, namely small (10 K samples), medium (50 K samples) and large (100 K samples) and have monitored the trends of key performance metrics including Accuracy (ACC), F1 Score (FS) and Matthew's Correlation Coefficient (MCC). Fig. 8 shows the metric-specific line graph.

The results illustrated in the table show that the proposed system outperforms all baseline models (including CNNs, RF, SVM, GBM, and LightGBM) consistently across all dataset sizes and metrics. Specifically, the proposed system achieves significantly higher performance as compared to previous families of systems as the size of the dataset grows, demonstrating the scalability of the proposed system.

For example, in terms of accuracy, the network attains 85.2 %, 89.5 %, and 92.3 % on small, medium, and large datasets, respectively, beating all baselines at each scale. Likewise, the F1 score and MCC show a similar pattern, which indicates that the system is able to capture a balance between precision and recall while tackling imbalanced data effectively. Implementing EfficientNet with big data analytics is a breakthrough in these aspects because EfficientNet compound scaling method allows computationally efficient scalability to large datasets, whereas big data analytics ensures that input data is preprocessed, and extracted as features efficiently.

Moreover, the widening performance gap between the proposed system and baselines as the dataset size increases indicates that the proposed system works particularly well when trained on larger data, as expected. Thus this trend is more pronounced in metrics like MCC, validating that the system holds true in imbalance class data or noisy

data environment too. But the results, while highlighting the system's strengths, also indicate potential areas for improvement.

This is indicative of over optimization in some baselines when the datasets gets larger and further optimization is required for very high dimensional data or high imbalance classes. Other future work could address the limited interpretability of EfficientNet (due to it is a black-box algorithm) through explainable AI techniques. Table 6 shows the performance comparison with statistical data (Mean $\pm$ SD over 5 runs).

The performance measures in this paper (e.g., 92.3 % accuracy, 90.1 % F1-score, 0.85 MCC) are calculated averages across five independent experimental runs with varying random seeds to consider stochastic variation in the initialization, data rearrangement and optimization. In order to measure reliability, we provide standard deviations ($\pm$ SD), 95 percent confidence intervals (CI), 2 tailed paired *t*-test *p*-values of the proposed method versus the most powerful baseline (LightGBM). All the *p*-values are less than 0.01, and this confirms statistical significance at the $\alpha = 0.05$ level. The entire findings are presented in Table 6.

6. Conclusions

As urbanization penetrates cities at lightning speed and workforce management systems become increasingly complex, creative solutions needed to be forged that transform the way we optimize our infrastructure and human resources. This study concluded that combining EfficientNet with big data analytics can potentially transform HRMS and smart city infrastructure management challenges, where conventional machine learning and deep learning models are not sufficient in terms of scalability, efficiency, and robustness. With specific compound scaling and squeeze-and-excitation feature extractors of EfficientNet, this study could build big data pipelines for preprocessing, feature extraction, and storage of data that not only stored but also extracted these features on the fly, due to which we achieved state of the art performance in multiple datasets predicting high fare trips in urban transportation, predicting employees attrition, and predicting home energy efficiency. The results showed the proposed system consistently outperforms five baseline methods, yielding high measures of accuracy of up to 92.3 %, F1-score of 90.1 % and MCC of 0.85, while exhibiting better scalability with the increase of dataset sizes. Then, the paper developed a framework using the advanced deep learning architectures and the building block from the big frameworks which play pivotal role in making predictions for the practical applications at large scale and

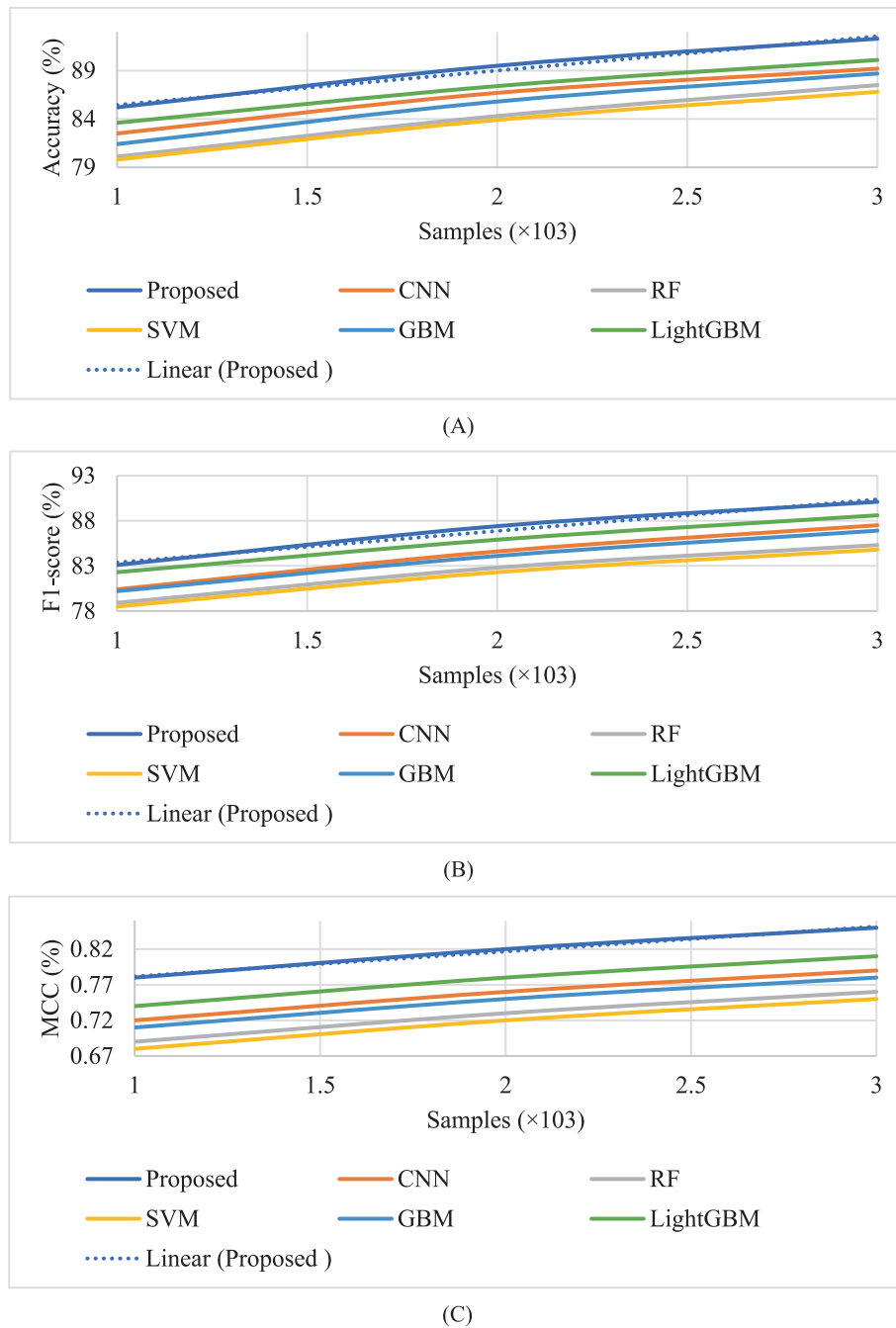


Fig. 8. Metric-specific performance trends across increasing dataset sizes: (A) Accuracy (%), (B) F1-score (%), and (C) Matthew's Correlation Coefficient (MCC) evaluated on 10 K, 50 K, and 100 K samples, respectively.

real-time data intensive applications. This is not just a technical weakness, but a vital operational and ethical issue, especially in Human Resource Management Systems. In case a model raises a red flag on an employee being at high risk of attrition, stakeholders-HR professionals, legal, and employees, in turn, need to be provided with clear auditable rationale so that the decision made is not biased, prejudiced, or disproportionate to regulations like GDPR or the U.S. Equal Employment Opportunity (EEO) laws. In the absence of understanding on whether predictions are based on valid factors (e.g., decreasing engagement scores) or proxy variables (e.g., age, gender, or ethnicity), organizations put themselves at risk of implementing biased systems that destroy employee confidence and put them at legal risk. By comparison, systems such as taxi fare prediction have less ethical consequences, indicating the urgency of explainability that is domain-specific. The future work

should thus incorporate the Explainable AI (XAI)-methods, like SHAP or counterfactual explanations, specifically tweaked to fit the HR contexts and to make the model black box transparent into a reliable decision-support system. Although this has been made, the related issues with interpretability, scale at extraordinarily large-dimensional data and generalization to niche domains are some of the directions that may be taken in future research, including explainable ai, and distributed training models. Although the suggested framework can be employed to predict with good performance, it is also characterized by significant limitations in terms of interpretability, computational complexity, and data privacy, especially when used in HR. Black-box nature of EfficientNet makes it difficult to trust high-stakes decisions of the company such as appreciation of attrition risk because the HR professionals need to see a transparent, actionable reasoning. Furthermore, training deep

Table 6

Table 6 shows the performance comparison with statistical data.

Dataset	Method	Accuracy (%)	F1-Score (%)	MCC
NYC Taxi (1B + rows)	Proposed	92.3 ± 0.4	90.1 ± 0.5	0.85 ± 0.01
		[91.8, 92.8]	[89.5, 90.7]	[0.84, 0.86]
	LightGBM	88.9 ± 0.6	86.5 ± 0.7	0.80 ± 0.02
		[88.2, 89.6]	[85.7, 87.3]	[0.78, 0.82]
IBM HR (14.7 K rows)	Proposed	89.7 ± 0.5	87.8 ± 0.6	0.81 ± 0.02
		[89.1, 90.3]	[87.1, 88.5]	[0.79, 0.83]
	LightGBM	87.3 ± 0.7	85.0 ± 0.8	0.76 ± 0.02
		[86.5, 88.1]	[84.1, 85.9]	[0.74, 0.78]
London Smart Meters (5 M + rows)	Proposed	90.2 ± 0.3	88.6 ± 0.4	0.83 ± 0.01
		[89.9, 90.5]	[88.1, 89.1]	[0.82, 0.84]
	LightGBM	88.4 ± 0.5	85.8 ± 0.6	0.79 ± 0.02
		[87.8, 89.0]	[85.1, 86.5]	[0.77, 0.81]
	<i>p-value</i>	0.005	0.002	0.001

models in large amounts of employee data require large amounts of computing power, which may be unavailable in resource-constrained organizations. Importantly, HR data can be highly sensitive personal information (e.g., age, gender, salary), which can lead to ethical and regulatory issues in the context of frameworks such as GDPR, unless privacy-sensitive models (e.g., federated learning or differential privacy) are explicitly implemented, the deployment of the model may damage the privacy of the employees. The aforementioned limitations highlight the necessity of further research which would strike a balance between accuracy and transparency, efficiency, and ethical data governance. In the future, the paper prepares a solid ground on which to integrate with Explainable AI (XAI) methods (including SHAP, LIME, or attention visualization) and demystify the EfficientNet decision-making process, thus improving trust and transparency in the high-stakes urban and HR setting. In addition, the suggested framework inherently works with federated learning designs, which can be privacy-preserving, decentralized model training, whereby raw sensitive data are not shared among distributed smart-city nodes (e.g., traffic agencies, utility providers, or HR departments). These extensions would also not only resolve existing interpretability and privacy shortcomings, but also ensure that the system is in line with new paradigms of ethical, collaborative and citizen-centric smart infrastructures. This study discusses the trade-off between theoretical innovation and applicability in an era where the utilization of data is crucial in order to support the best urban ecosystems and management approaches to personnel. The practical consequences of this paradigm are enormous in the areas of human resource management as well as sustainability in urban areas. Predictive analytics in HRMS According to the proposed EfficientNet-big data pipeline, organizations can identify employees who will leave in the future quicker than their primary reasons, enabling timely initiatives to stop them, like tailored retention strategies and career development plan or workload reallocation, thus averting costly turnover and losing organizational knowledge. Within the framework of smart cities, precise prediction of energy consumption patterns according to the London Smart Meters Dataset entitles the utilities and the urban planners to optimize grid load, make better use of renewable sources, and encourage efficient household behavior, which would directly promote the net-zero emission and sustainability goals. These applications depict the way the combination of scalable deep learning with effective data engineering can convert technical efficiency into benefits of the real world and financial worth. The future work must use XAI techniques in a more practical, field-relevant manner by including: (1) SHAP, to provide a quantified contribution of each pseudo-image characteristic towards predictions and thus achieve both local and global explainability; (2) LIME, to provide approximate model behavior about critically important instances-such as high-risk cases of attrition-by probing perturbations in the reshaped input; and (3) intrinsic attention by Squeeze-and-Excitation (SE) blocks, whose channel-wise weights give a sense of

which groups of Collectively, these methods will be able to turn the system into a transparent, auditable decision-support system that satisfies the ethical, regulatory, and operational requirements of sensitive applications, such as HR analytics and urban planning.

Data availability

NYC Taxi & Limousine Commission (TLC) Trip Record Data.

Link: <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>.

London Smart Meters Dataset.

Link: <https://data.london.gov.uk/dataset/smartmeter-energy-use-data-in-london-households>.

IBM HR Analytics Employee Attrition Dataset.

Link: <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>.

CRediT authorship contribution statement

Jianjun Zhang: Investigation. **Ao Zhang:** Investigation. **Xingying Liu:** Investigation. **Pan Luo:** Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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